ZHANG JIANNAN

Flight Control Engineer & PhD Candidate

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jiannan-zhang

STRENGTHS

- Flight Control System
- Flight Physics Modelling
- MBSE
- Certification
- Flight Testing

SOFTWARE SKILLS

- MATLAB/Simulink
- Control Design Toolbox
- Math Symbolic Toolbox
- Git
- C/C++

LANGUAGES

English: Professional
Chinese: Native
German: Goethe B1

SUMMARY

Motivation: Productize eVTOLs.
Industry: 10+ years on eVTOL FCS, 1 TC, 1000+h flight testing.
Academy: 10+ publications, 300+ citations.

PROFESSIONAL EXPERIENCE

Research Associate | Institute of Flight System Dynamics, TUM

06.2016 – Now | Munich, Germany

- Lead of control allocation algorithm.
- Lead of control configuration optimization.
- Flight control system development.
- Flight dynamics modelling toolchain development.

Lead of Flight Control System | AutoFlight

08.2023 – 07.2024 | Shanghai, China

- Lead of GNC function development.
- Lead of system architecturing.
- Lead of FCS TC applications.

Senior Flight Control Engineer | AutoFlight

07.2021 – 08.2023 | Shanghai, China

- Lead of guidance and control law development.
- Lead of flight dynamics modelling development.
- System architecturing.
- Certification tasks (requirement capture and validation).
- Flight testing.

Flight Control Engineer | Autel Robotics Europe GmbH

06.2016 – 01.2020 | Ismaning, Germany

- Flight control algorithm for tilt-rotor UAV (Dragonfish).
- Attitude loop design and control allocation.
- Flight testing.

Software Developer | Phoenix-Wings GmbH

06.2020 – 07.2021 | Ismaning, Germany

- Lead of trim and linearization tool.
- Simulation toolchains development.
- Flight testing.
- Project management.

EDUCATION

Doctoral Candidate Technical University of Munich (TUM)

11.2017 – Now (Expected Grad.: 2026) Munich, Germany

- **Research Direction**: eVTOL Control Authority and Allocation Optimization

M.Sc. Aerospace Engineering Nanyang Technological University (NTU)

08.2013 – 02.2016 Singapore

- **CGPA**: 4.92 (CAP: 1.06) **Major Subjects**: Flight Control, Light Weight Structures

B.Eng. Aircraft Design Engineering Beijing Institute of Technology (BIT)

08.2008 – 07.2012 Beijing, China

- **GPA**: 85/100 **Major Subjects**: Flight Mechanics, Aerodynamics

INDUSTRY PROJECTS

CarryAll - 2000kg Lift-cruise eVTOL | Autoflight

07.2021 – 08.2024 Industry Project

- **Role**: Lead of modelling, guidance and control law
- **Contribution**: Control law design, SW implementation, system architecturing and certification tasks (FHA, PSSA, etc.).
- **Highlight**: First 2000kg eVTOL TC in the world.

V280 - 280kg Lift-cruise eVTOL | TUM-FSD

02.2020 – 06.2021 Industry Project

- **Role**: Lead of control allocation development
- **Contribution**: Control allocation algorithms for hover, transition and cruise flight, fail-safe proven for OEI.
- **Highlight**: Developed under code-compliant guidelines specified by DO-331 (standard libraries, Model Advisor, Test Harnesses, etc.).

Manta Ray - 30kg Lift-cruise eVTOL | Phoenix-Wings GmbH

06.2020 – 07.2021 Industry Project

- **Role**: Lead of Trim and Linearization Tool in C++
- **Contribution**: Implemented the trim and linearization tool in C++, and the MATLAB/Simulink interface for model verification and further design tasks.

Dragonfish - 7kg Tilt-rotor UAV | Autel Robotics Europe GmbH

06.2016 – 01.2020 Industry Project

- **Role**: Lead of control allocation development
- **Contribution**: Develope nonlieanr control allocation for the entire flight envelope. Support INDI attitude and velocity loop design. Two first-author papers published.
- **Highlight**: Dragonfish as a product has already generated billions of sales revenues.

RESEARCH PROJECTS

Generalized Control Allocation Optimization & Protection | TUM-FSD

01.2025 – 12.2025 Research Project

- **Role**: Lead of algorithm development
- **Contribution**: An adaptive method which generalize p-norm optimization tagets. Two command protection methods to improve pseudinverse based methods. One first-author and two second-author journals published.

eVTOL Control Configuration Optimization | TUM-FSD

05.2020 – 12.2024 Research Project

- **Role**: Lead of algorithm design and SW implementation
- **Contribution**: Developed a method to optimize eVTOL powered-lift rotor topology. The optimal configuration features sublime robustness against OEI scenario. Two first-author journals published.

F-16 Modelling Extension and Control Design | TUM-FSD

05.2016 – 01.2020 Research Project

- **Role**: Lead of modelling and INDI baseline controller
- **Contribution**: Developed the INDI closed-loop simulation framework and the nonlinear optimal control allocation and advanced hedging algorithm. Two first-author papers published.

PUBLICATIONS

[1] Zhidong Lu, **Jiannan Zhang**, Hangxu Li, and Florian Holzapfel. "A Virtual Command Protection Strategy for Pseudo-Inverse Flight Control Allocation". In: *Aerospace Science and Technology* (2025), p. 111134.

[2] Zhidong Lu, **Jiannan Zhang**, Hangxu Li, and Florian Holzapfel. "An Adaptive Command Scaling Method for Incremental Flight Control Allocation". In: *Actuators*. Multidisciplinary Digital Publishing Institute. 2025.

[3] **Jiannan Zhang**, Max Söpper, Florian Holzapfel, and Shuguang Zhang. "Four-Dimensional Generalized AMS Optimization Considering Critical Engine Inoperative for an eVTOL". in: *Aerospace* 11.12 (2024), p. 990.

[4] **Jiannan Zhang** and Florian Holzapfel. "Saturation Protection with Pseudo Control Hedging: A Control Allocation Perspective". In: *AIAA Scitech 2021 Forum*. 2021, p. 0370.

[5] **Jiannan Zhang**, Max Söpper, and Florian Holzapfel. "Attainable moment set optimization to support configuration design: a required moment set based approach". In: *Applied Sciences* 11.8 (2021), p. 3685.

[6] Pranav Bhardwaj, Stefan A Raab, **Jiannan Zhang**, and Florian Holzapfel. "Thrust command based integrated reference model with envelope protections for tilt-rotor vtol transition uav". In: *AIAA Aviation 2019 Forum*. 2019, p. 3266.

[7] Stefan A Raab, **Jiannan Zhang**, Pranav Bhardwaj, and Florian Holzapfel. "Consideration of control effector dynamics and saturations in an extended INDI approach". In: *AIAA Aviation 2019 forum*. 2019, p. 3267.

[8] **Jiannan Zhang**, Pranav Bhardwaj, Stefan A Raab, and Florian Holzapfel. "Control allocation framework with SVD-based protection for a tilt-rotor VTOL transition air vehicle". In: *AIAA Aviation 2019 Forum*. 2019, p. 3265.

[9] **Jiannan Zhang**, Jian Wang, Fubiao Zhang, and Florian Holzapfel. "Modeling and incremental nonlinear dynamic inversion control for a highly redundant flight system". In: *AIAA Scitech 2019 Forum*. 2019, p. 1922.

[10] Pranav Bhardwaj, Stefan A Raab, **Jiannan Zhang**, and Florian Holzapfel. "Integrated reference model for a tilt-rotor vertical take-off and landing transition uav". In: *2018 Applied Aerodynamics Conference*. 2018, p. 3479.

[11] Stefan A Raab, **Jiannan Zhang**, Pranav Bhardwaj, and Florian Holzapfel. "Proposal of a unified control strategy for vertical take-off and landing transition aircraft configurations". In: *2018 Applied Aerodynamics Conference*. 2018, p. 3478.

[12] **Jiannan Zhang**, Pranav Bhardwaj, Stefan A Raab, Saurabh Saboo, and Florian Holzapfel. "Control allocation framework for a tilt-rotor vertical take-off and landing transition aircraft configuration". In: *2018 Applied Aerodynamics Conference*. 2018, p. 3480.